

Chapter 8 Homework: Vector Norms

Part A - Python Skills

- Create vectors and compute L1 norms using `numpy.linalg.norm(..., 1)`.
- Compute L2 norms using `numpy.linalg.norm(...)`.
- Compute max norms using `numpy.linalg.norm(..., np.inf)`.
- Compare norms for several vectors.
- Write your own functions that manually compute L1 and L2 norms.

Part B - Practical Problem

A sensor system measures three threat indicators: [Range Score, Velocity Score, Intent Score]

Represent each target as a vector. Target A = [8, 2, 1] Target B = [4, 5, 4] Target C = [2, 8, 7]

Tasks: 1. Compute the L1, L2, and Max norms of each target. 2. Rank the targets under each norm.

3. Explain why different norms produce different rankings. 4. Discuss which norm might be most appropriate if large values in any single dimension are especially concerning.